DNA and RNA

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Intended Learning Outcomes

1. Identify the three core structural components of a nucleotide monomer and explain how minor chemical variations in pentose sugars differentiate DNA from RNA.
2. Apply the principles of the Central Dogma of Molecular Biology to trace the flow of genetic instruction from DNA to RNA to functional protein.
3. Evaluate the structural and operational differences between DNA and RNA
4. Describe the distinct structural features and diverse physiological functions of specialized RNA classes

Deoxyribonucleic Acid (DNA) and **Ribonucleic Acid (RNA)**

These are the two main types of **nucleic acids**, and they are responsible for

- storing
- transmitting
- and expressing the genetic information that defines an organism.

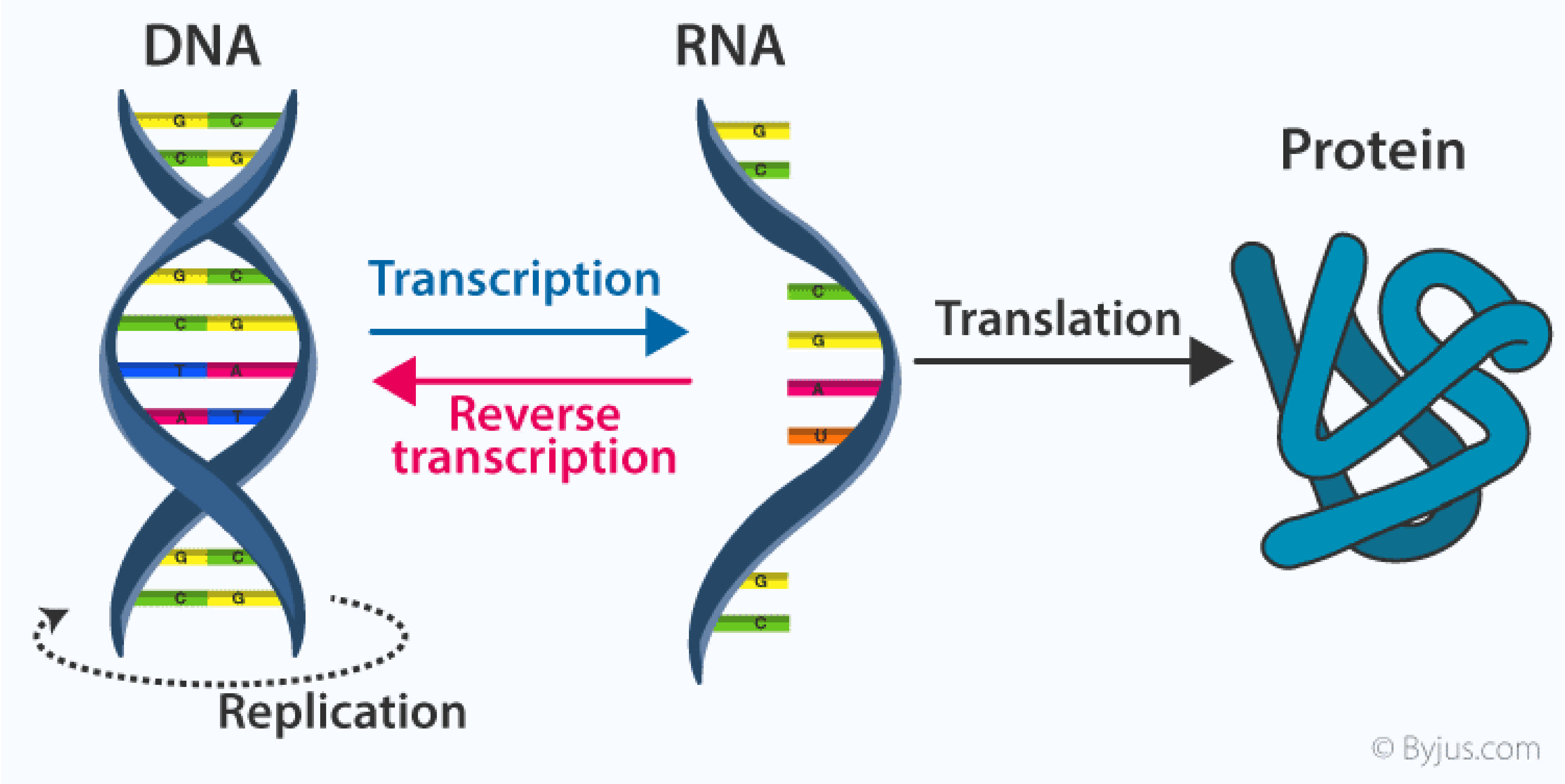
Central Dogma of Life

- The relationship between DNA and RNA is at the heart of what's known as the **Central Dogma of Molecular Biology**:

DNA → RNA → Protein

- In this process, DNA serves as the master blueprint, RNA acts as the versatile messenger and a key part of the cellular machinery, and proteins are the functional workhorses of the cell.

- Understanding the structure and function of DNA and RNA is fundamental to understanding all life.



DNA
RNA
Protein
Transcription
Reverse transcription
Translation
Replication
G C
C G
G C
C G
T A
A T
G C
C G
G
C
C
G
A
U
G
C
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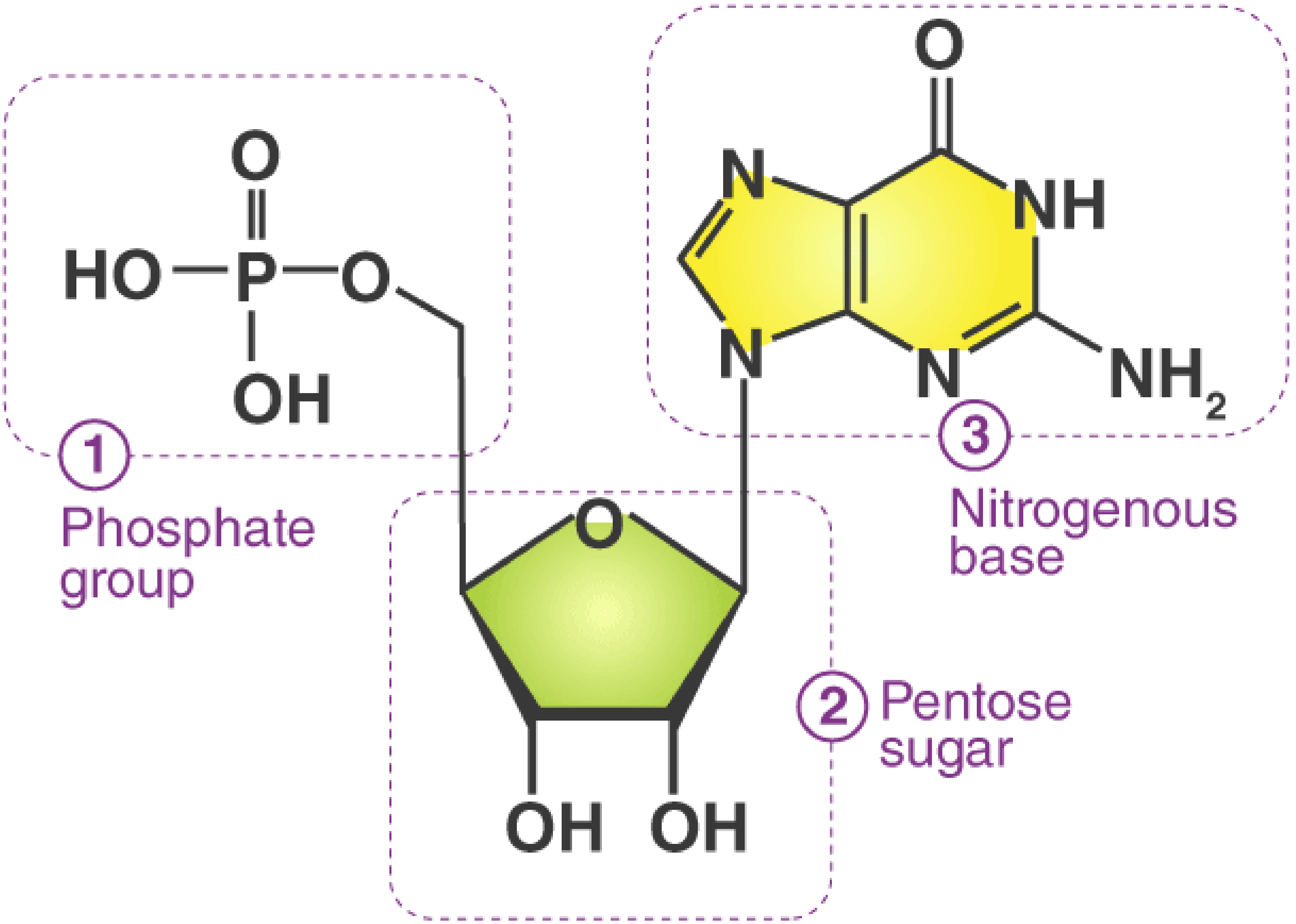
The Monomeric Unit: Nucleotides

- Both DNA and RNA are polymers, meaning they are long chains made up of repeating single units. The monomer for both nucleic acids is a **nucleotide**.

A nucleotide has three main components:

- **A Pentose Sugar**
- **A Nitrogenous Base**
- **One or More Phosphate Groups**

Nucleotides with one, two, or three phosphates are called monophosphates, diphosphates, and triphosphates, respectively (e.g., ATP, ADP, AMP).

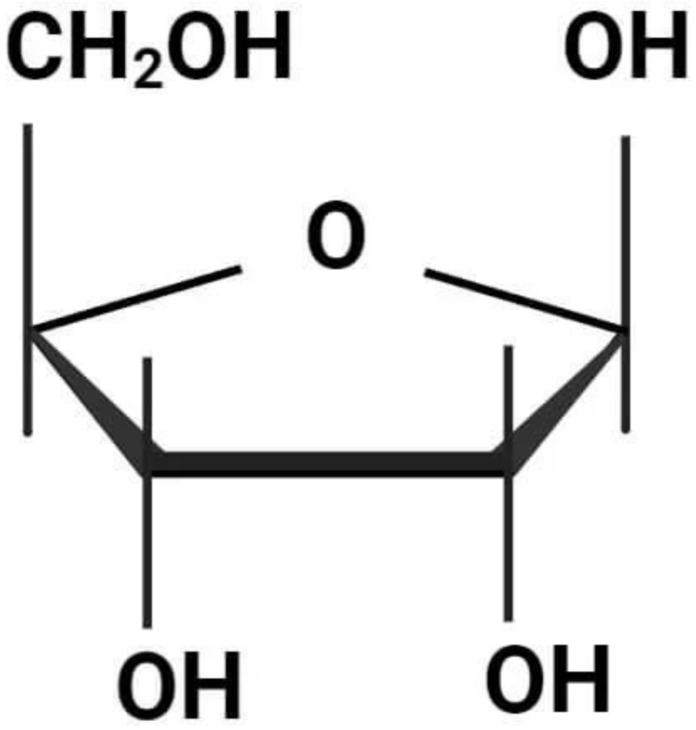

O
HO
P
O
OH
1
Phosphate group
O
OH
OH
2
Pentose sugar
O
N
NH
N
N
NH2
3
Nitrogenous base

A Pentose Sugar

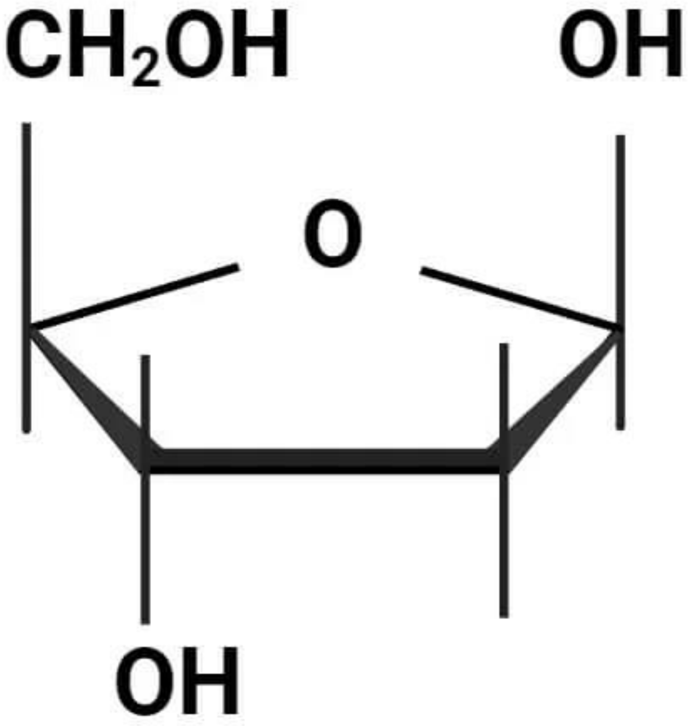
A five-carbon sugar.

- **In DNA:** The sugar is **2'-deoxyribose**. It is "deoxy" because it is missing a hydroxyl (-OH) group on the **2' carbon** of the ring.

- **In RNA:** The sugar is **ribose**. It has a hydroxyl (-OH) group on the **2' carbon**. This seemingly small difference is crucial for the stability and function of each molecule.



Ribose



2-deoxyribose

A Nitrogenous Base

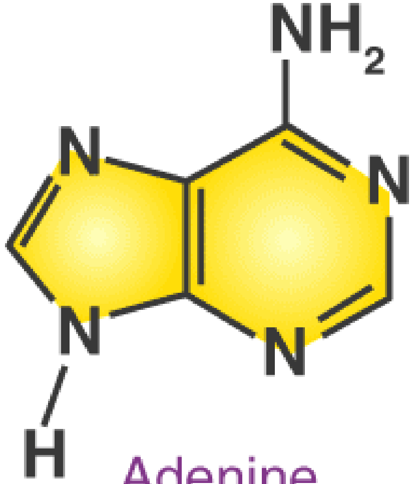
A nitrogen-containing ring structure. There are two main types:

Purines: Double-ring structures.

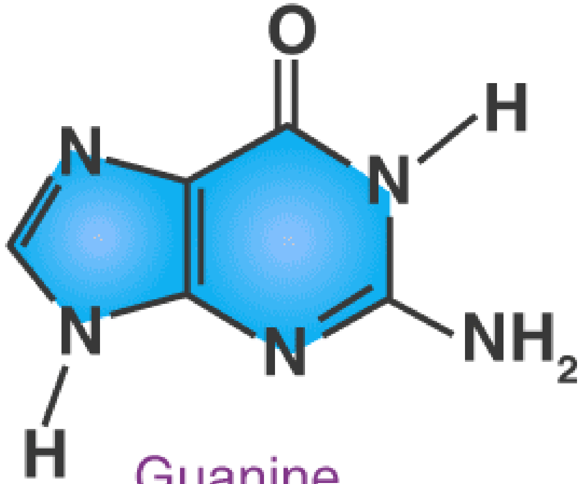
- **Adenine (A)**
- **Guanine (G)**

Pyrimidines: Single-ring structures.

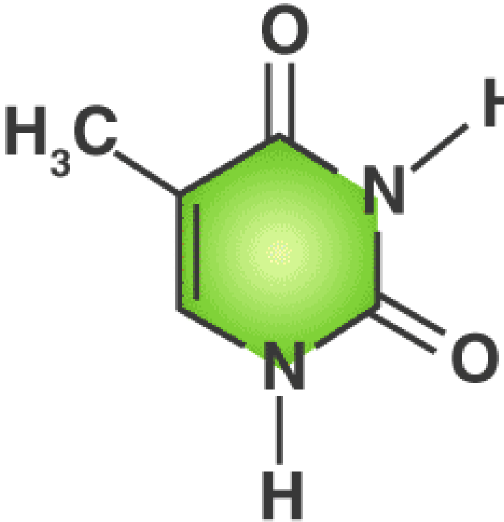
- **Cytosine (C)**
- **Thymine (T)** (found only in **DNA**)
- **Uracil (U)** (found only in **RNA**)



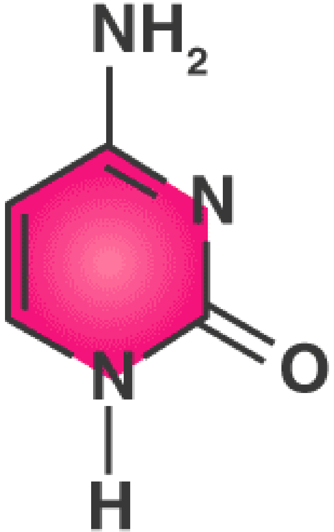
Adenine



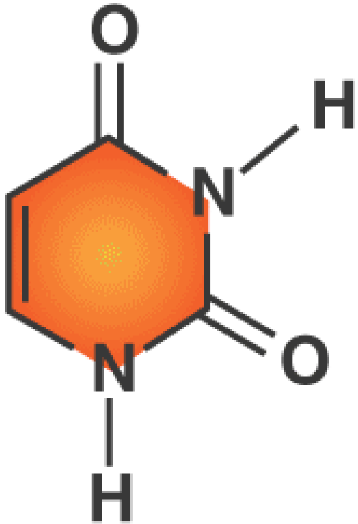
Guanine



Thymine



Cytosine



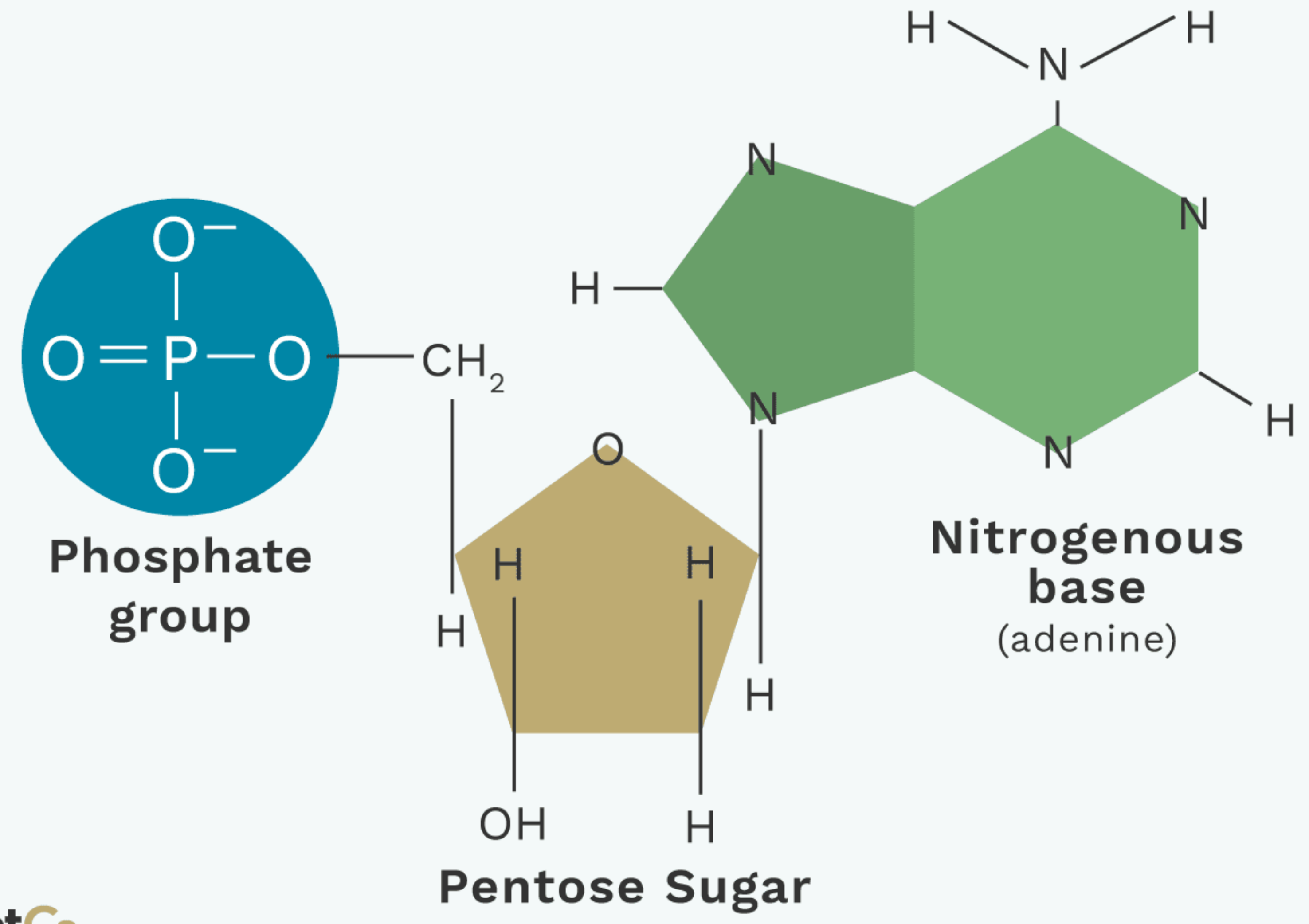
Uracil

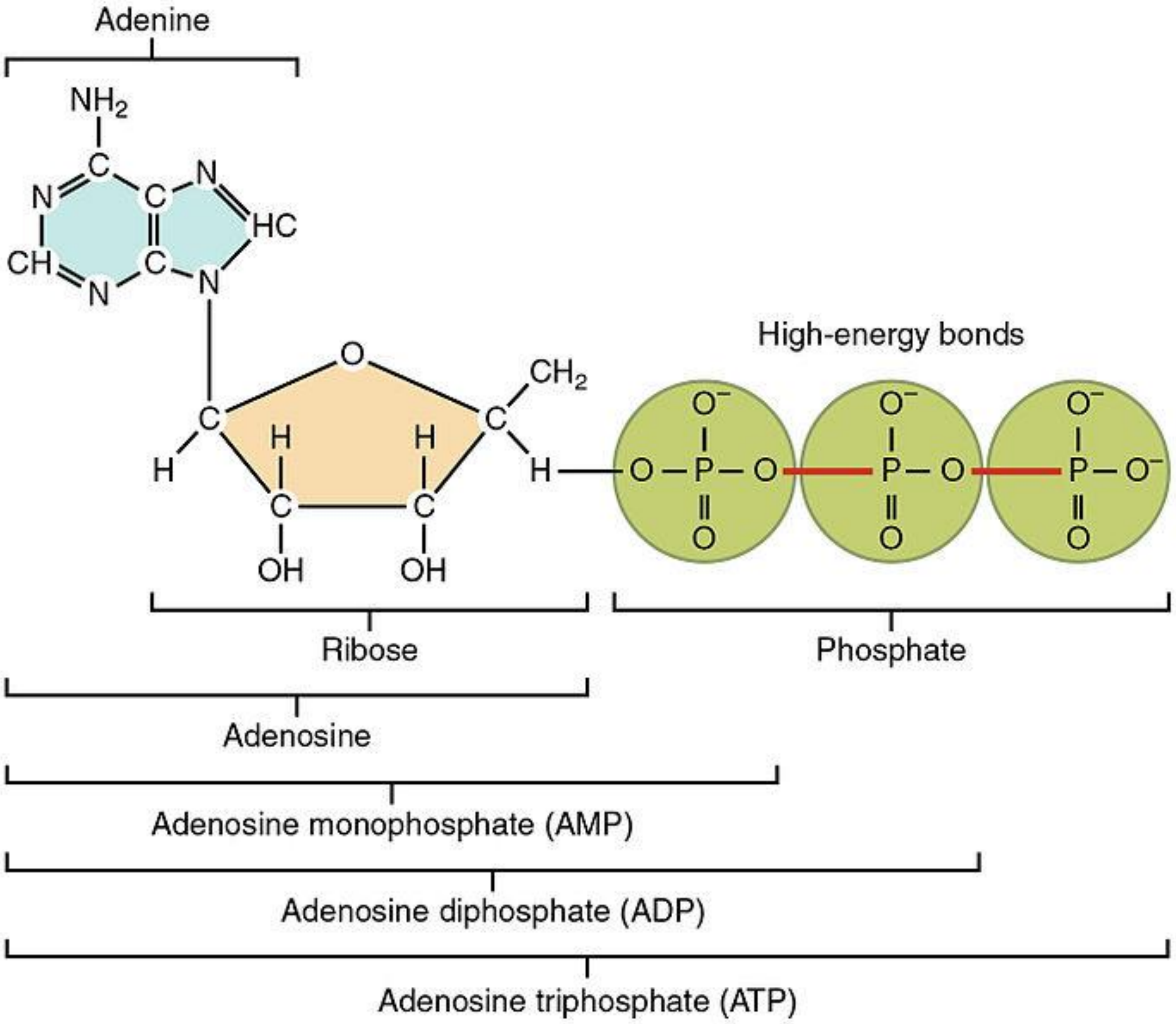
One or More Phosphate Groups

- Attached to the **5' carbon** of the sugar.

- Nucleotides with one, two, or three phosphates are called monophosphates, diphosphates, and triphosphates, respectively (e.g., ATP, ADP, AMP).

3 Parts of a Nucleotide




Adenine
High-energy bonds
Ribose
Phosphate
Adenosine
Adenosine monophosphate (AMP)
Adenosine diphosphate (ADP)
Adenosine triphosphate (ATP)

Naming Convention

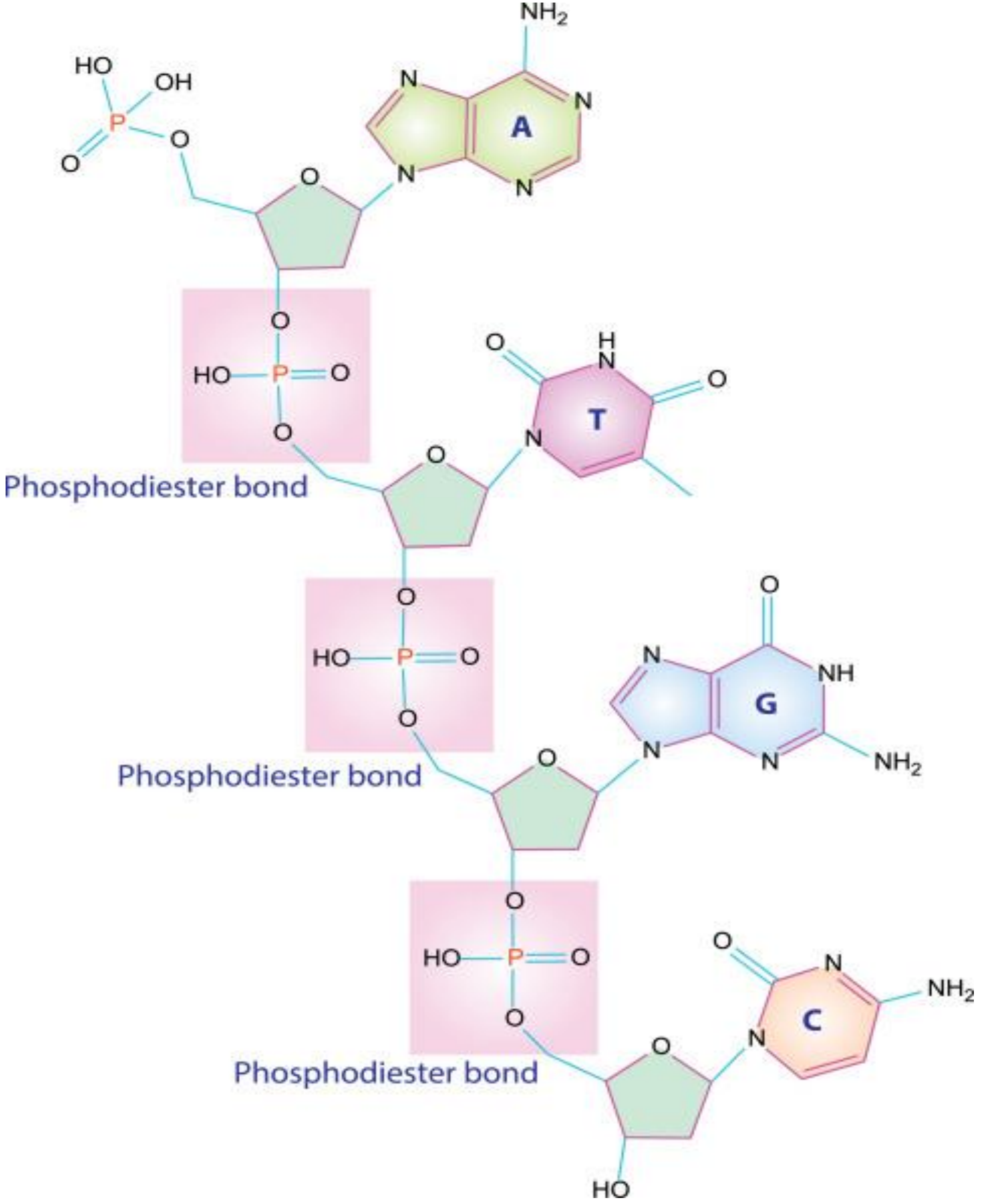
- A **nucleoside** is just the sugar + base (e.g., Adenosine).
- A **nucleotide** is the sugar + base + phosphate group(s) (e.g., Adenosine Monophosphate, AMP).

The Polymeric Structure: Nucleic Acid Chains

- Nucleotides link together to form a long polymer chain.

Phosphodiester Bonds: The individual nucleotides are joined by strong, covalent bonds called **phosphodiester bonds**.

- This bond forms between the **5'-phosphate group** of one nucleotide and the **3'-hydroxyl group** of the sugar of the next nucleotide.
- This linkage creates a repeating **sugar-phosphate backbone** that gives the nucleic acid its structural integrity.



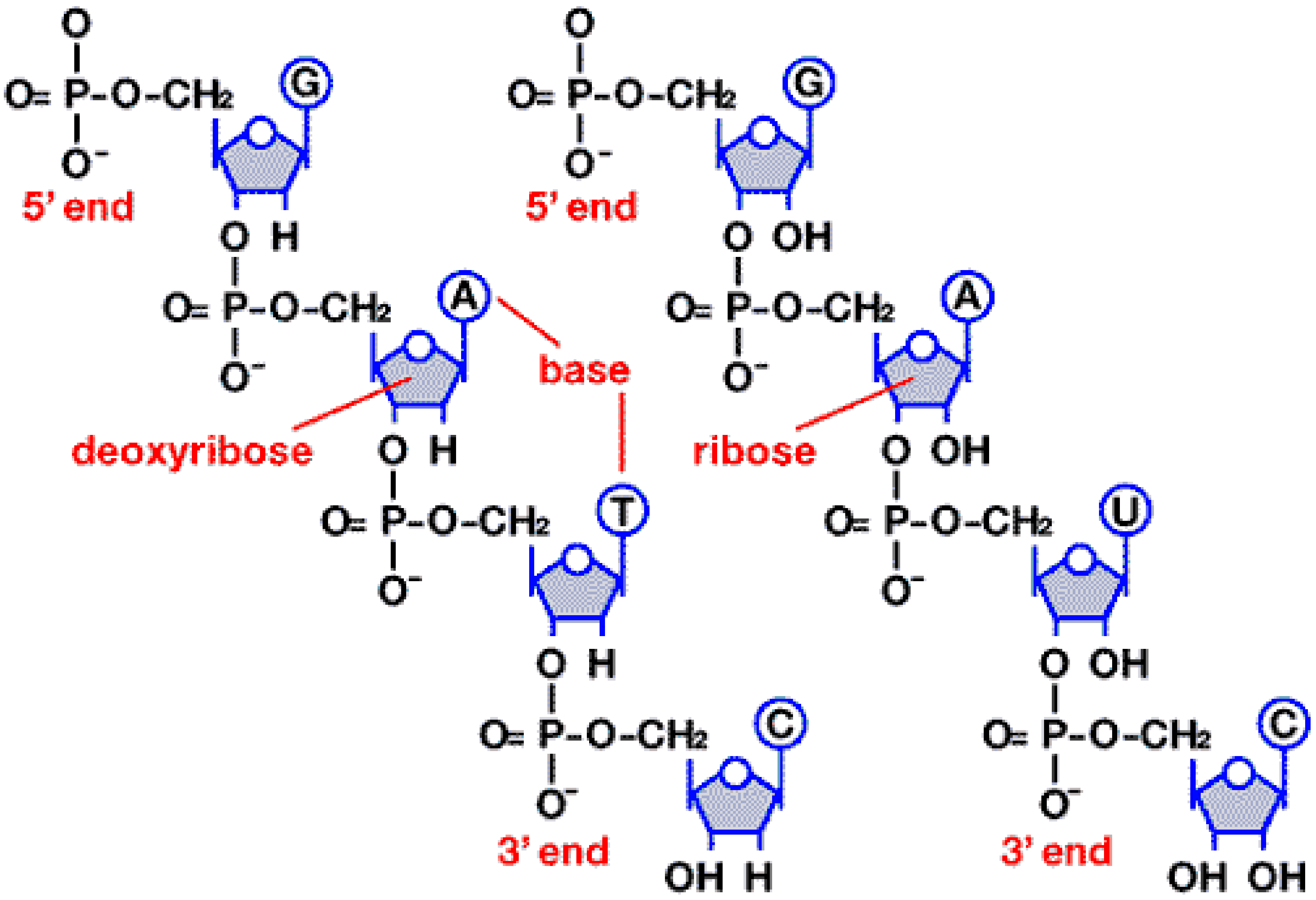
Phosphodiester bond
Phosphodiester bond
Phosphodiester bond

The Polymeric Structure: Nucleic Acid Chains

Directionality: A nucleic acid chain is not symmetrical; it has a distinct direction, or polarity.

- The end with a free phosphate group on the 5' carbon is called the **5' end**.
- The end with a free hydroxyl group on the 3' carbon is called the **3' end**.
- By convention, a nucleic acid sequence is always written from **5' to 3'**. This directionality is fundamental to how both DNA and RNA are synthesized and read.

Fig: H



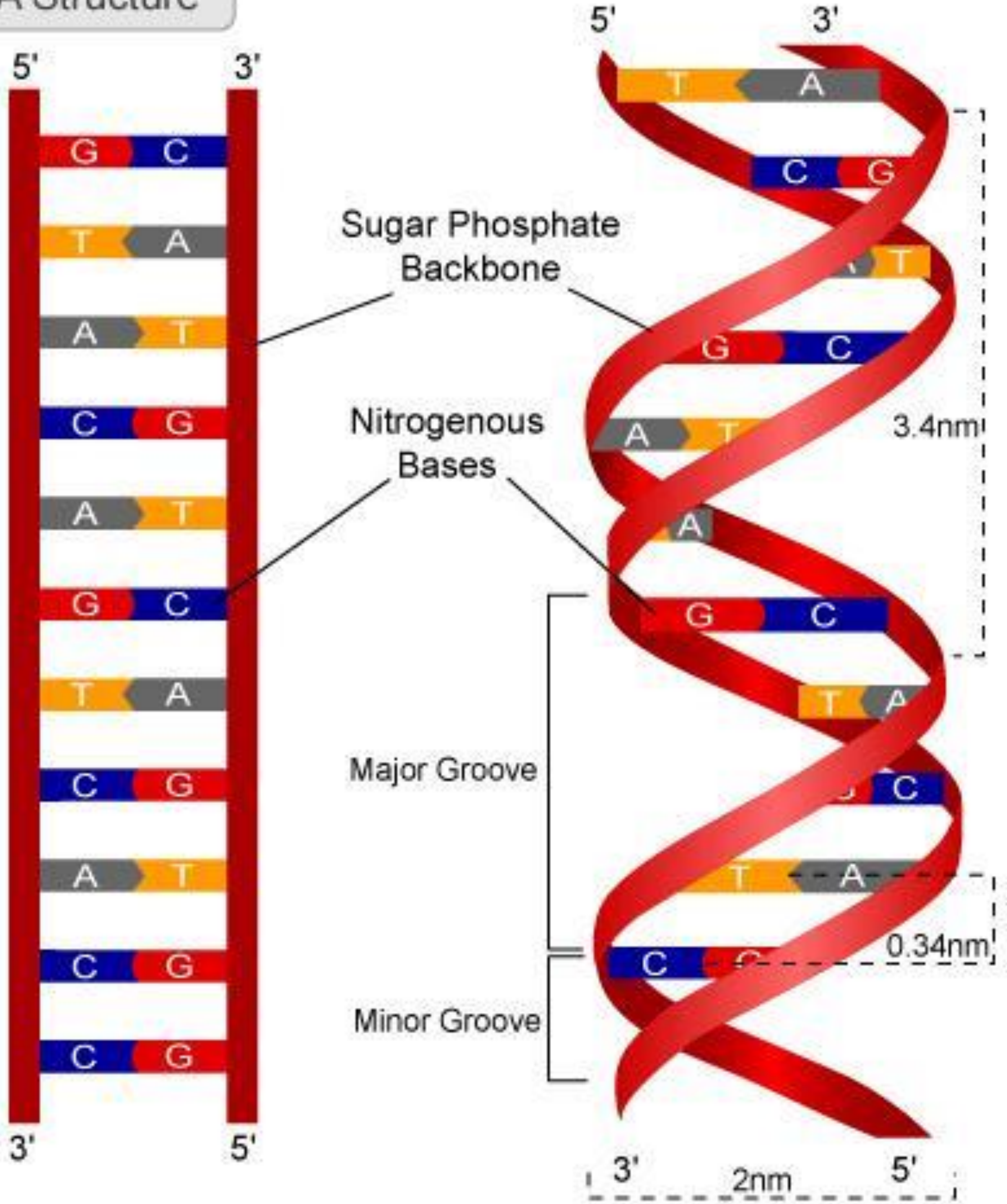
DNA: Structure

DNA's structure is one of the most famous discoveries in biology, elucidated by James Watson and Francis Crick in 1953.

A. The Double Helix:

- DNA exists as a **double helix**, a twisted ladder-like structure with two strands intertwined around each other.
- The two strands are **antiparallel**; they run in opposite directions. If one strand runs 5' to 3', the other runs 3' to 5’.
- The **sugar-phosphate backbones** are on the outside of the helix, forming the "sides" of the ladder.
- The **nitrogenous bases** are on the inside, stacked horizontally like the "rungs" of the ladder.

DNA Structure



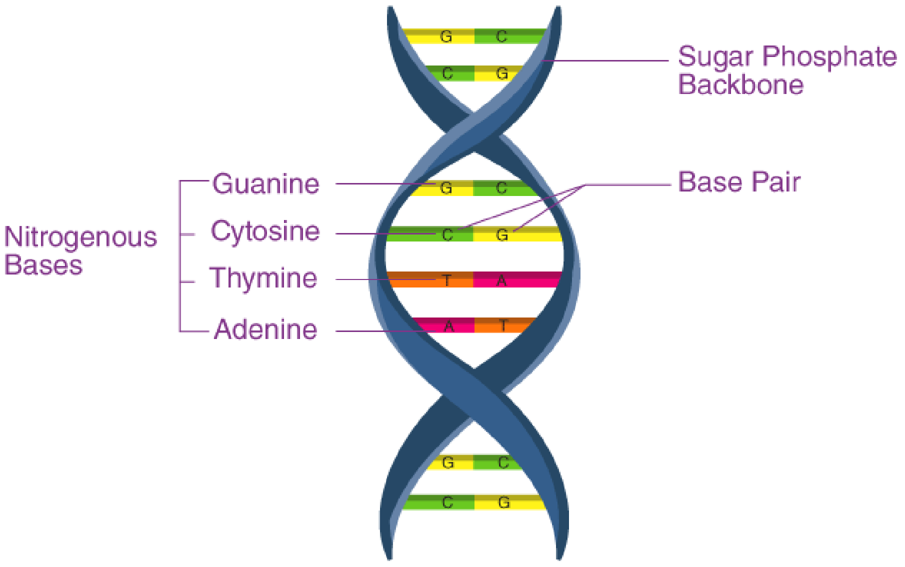
DNA: Structure

B. Base Pairing (Hydrogen Bonding):

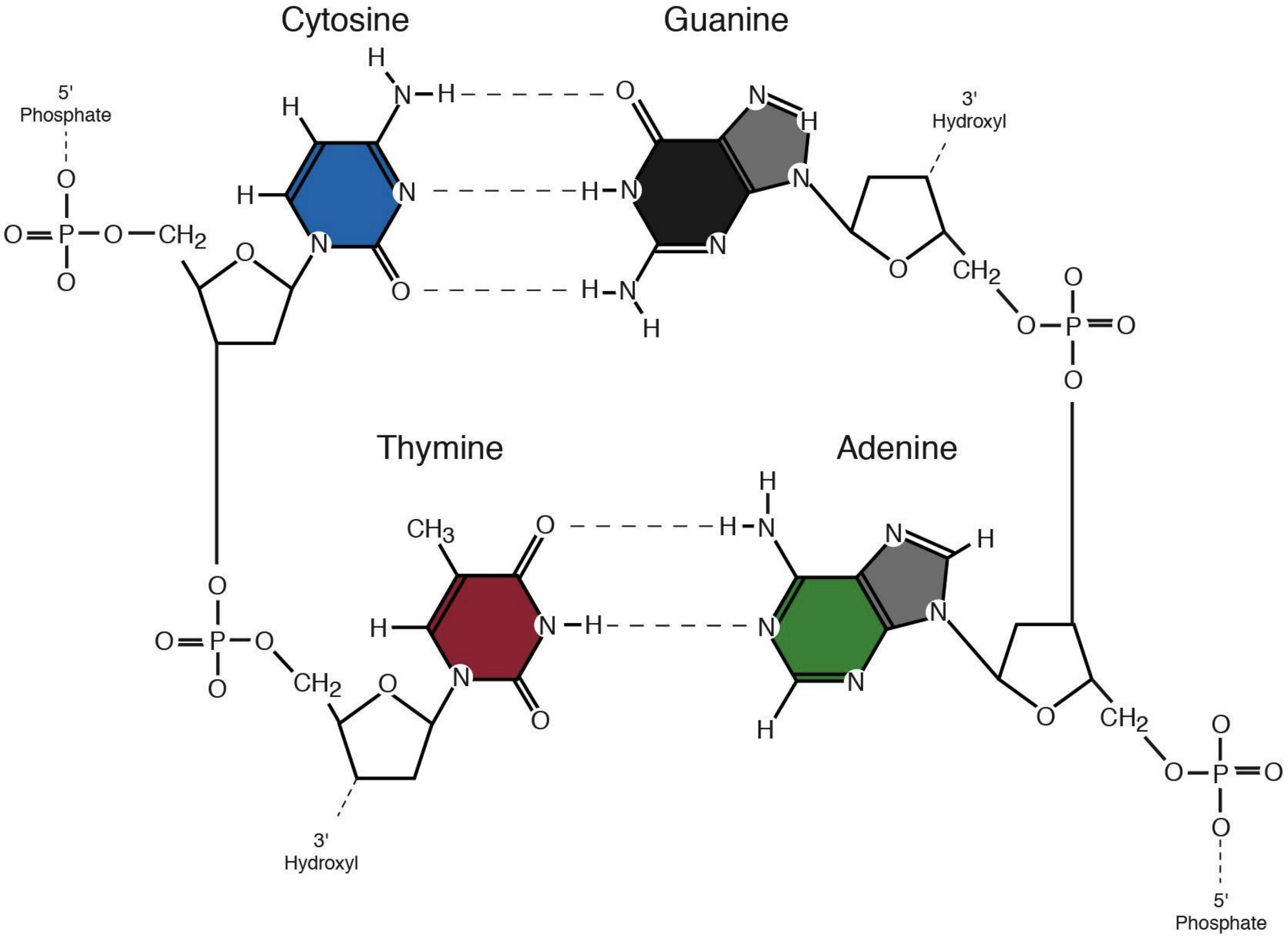
- The two strands of the double helix are held together by **hydrogen bonds** between the nitrogenous bases on opposite strands. This is a very specific process known as **complementary base pairing**.

- **Adenine (A) always pairs with Thymine (T),** forming **two** hydrogen bonds.
- **Guanine (G) always pairs with Cytosine (C),** forming **three** hydrogen bonds.

- This predictable pairing rule, first observed by Erwin Chargaff (A=T and G=C), is the key to DNA replication and repair.
- The G-C pair is slightly stronger and more stable than the A-T pair due to the additional hydrogen bond.



Sugar Phosphate Backbone
Base Pair
Nitrogenous Bases
Guanine
Cytosine
Thymine
Adenine



Cytosine
Guanine
Thymine
Adenine
5' Phosphate
3' Hydroxyl
3' Hydroxyl
5' Phosphate

Features of DNA

- **Genetic Information Storage:** The sequence of the four bases (A, T, C, G) along the DNA strand constitutes the genetic code. This sequence contains all the instructions needed to build and maintain an organism.

- **Stability:** The double-stranded structure, the strong sugar-phosphate backbone, and the protective helix shape make DNA extremely stable. This is essential for a molecule whose primary role is to be a permanent, long-term archive of genetic information.

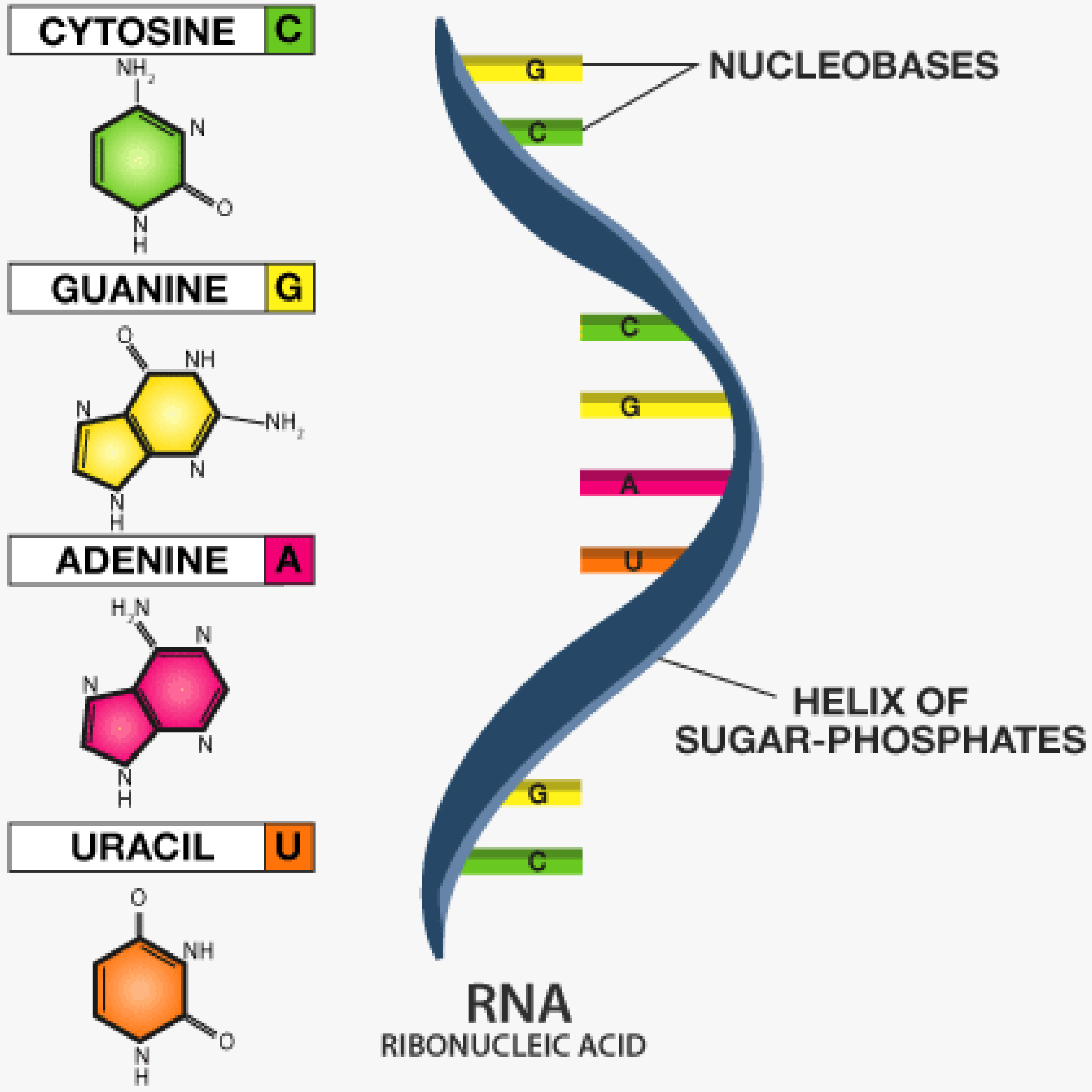
- **Replication:** The complementary base pairing allows the double helix to be unwound, and each original strand can serve as a template to synthesize a new, perfectly matched complementary strand. This ensures that genetic information is passed accurately from one generation of cells to the next.

RNA: Structure

The Single Strand:

- RNA is typically a **single-stranded** polymer.

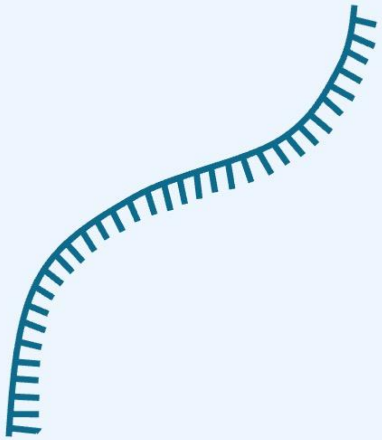
- However, because it's single-stranded, it can fold back on itself and form intricate three-dimensional structures through **intramolecular hydrogen bonding.** This folding is critical for its function.


CYTOSINE C
GUANINE G
ADENINE A
URACIL U
NUCLEOBASES
HELIX OF
SUGAR-PHOSPHATES
RNA
RIBONUCLEIC ACID

Key Structural Differences from DNA

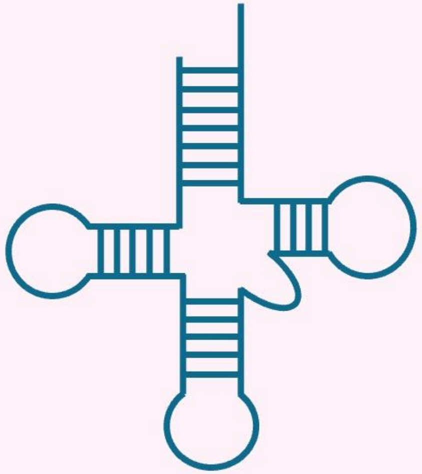
- **Sugar:** RNA contains **ribose** sugar (with a 2'-OH group). This makes RNA more chemically reactive and less stable than DNA. The 2'-OH group is a potential site for chemical attack, leading to hydrolysis of the phosphodiester backbone.
- **Base:** RNA contains **Uracil (U)** instead of Thymine (T). Uracil also pairs with Adenine (A).
- **Structure:** RNA is mostly single-stranded, while DNA is a double helix.

mRNA



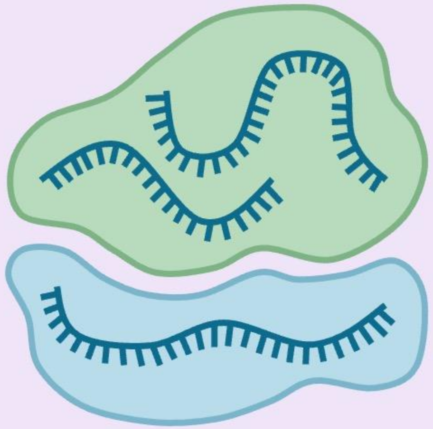
Encodes proteins

tRNA



Carries amino acids

rRNA

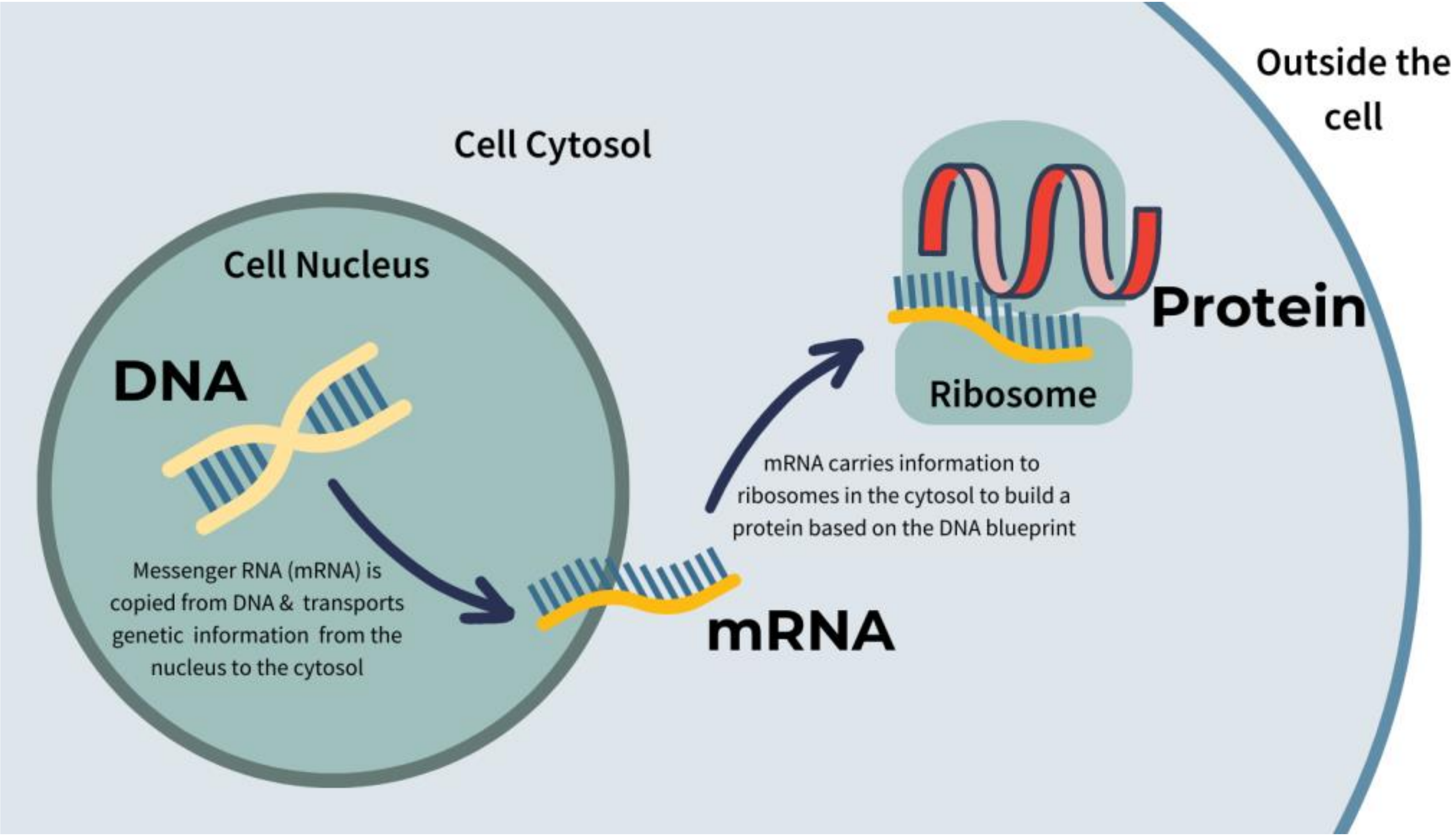


Forms the ribosome

Diverse Functions of RNA

Messenger RNA (mRNA)

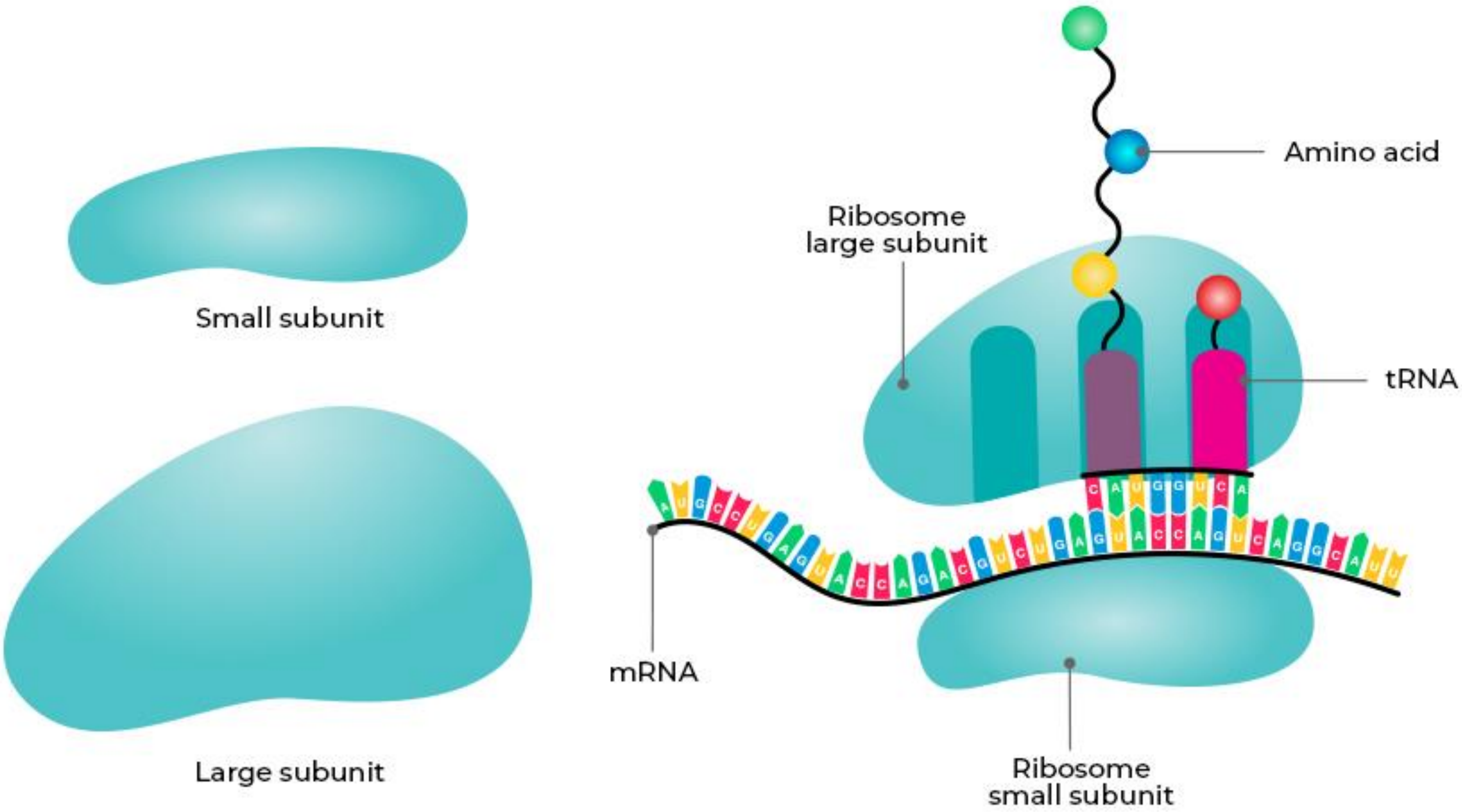
- mRNA carries the genetic code from DNA in the nucleus to the ribosome in the cytoplasm.
- The sequence of bases in mRNA is read in three-base segments called **codons**, each specifying a particular amino acid.


Outside the cell
Cell Cytosol
Protein
Ribosome
Cell Nucleus
DNA
mRNA carries information to ribosomes in the cytosol to build a protein based on the DNA blueprint
Messenger RNA (mRNA) is copied from DNA & transports genetic information from the nucleus to the cytosol
mRNA

Diverse Functions of RNA

Ribosomal RNA (rRNA)

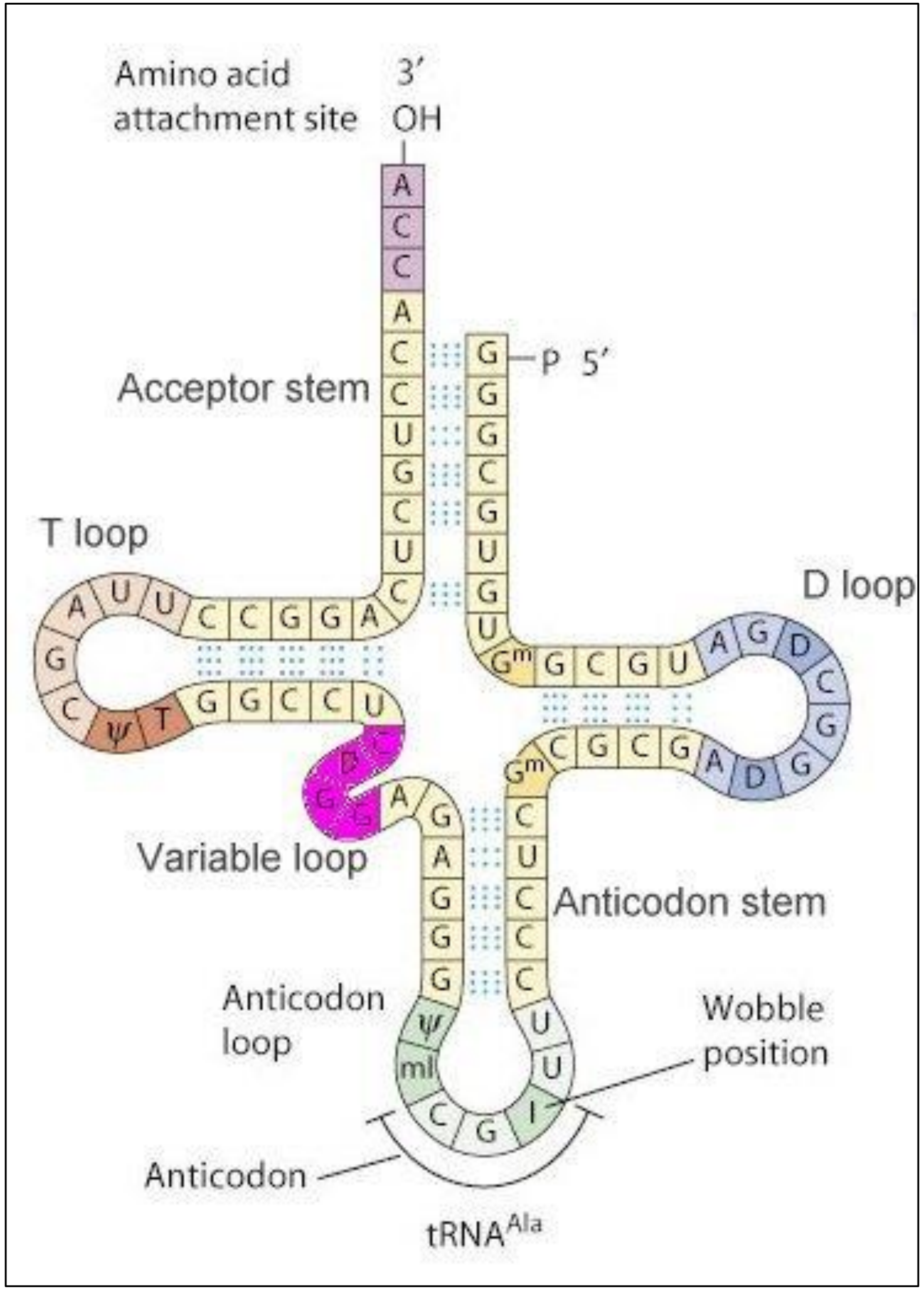
- rRNA is a major structural and catalytic component of **ribosomes**, the cellular machinery that synthesizes proteins.
- Importantly, a portion of rRNA has **catalytic activity** itself, acting as a **ribozyme** to form peptide bonds.

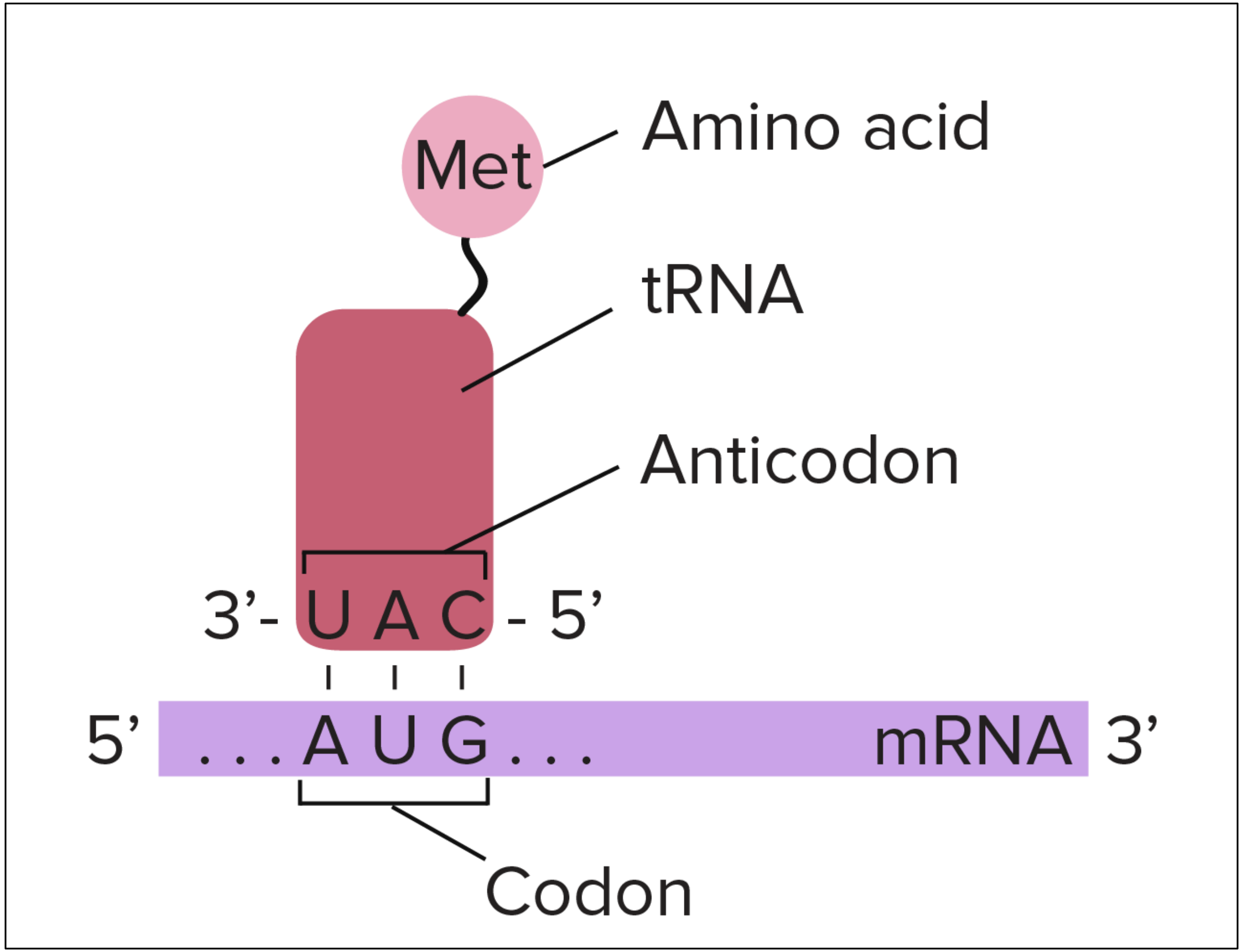

Small subunit
Large subunit
Amino acid
Ribosome
large subunit
tRNA
mRNA
Ribosome
small subunit

Diverse Functions of RNA

Transfer RNA (tRNA)

- tRNA acts as an **adaptor molecule**.
- Each tRNA molecule is "charged" with a specific amino acid and has a three-base **anticodon** that recognizes and pairs with a complementary codon on the mRNA.
- This ensures that the correct amino acid is brought to the ribosome at the right time.


Amino acid attachment site
3′ OH
Acceptor stem
P 5′
T loop
D loop
Variable loop
Anticodon stem
Anticodon loop
Wobble position
Anticodon
tRNA^Ala


Met
Amino acid
tRNA
Anticodon
3’- U A C - 5’
5’ . . . A U G . . . mRNA 3’
Codon

Diverse Functions of RNA

Other RNA Molecules

There are many other types of RNA with regulatory and catalytic functions, including:

- **Small nuclear RNA (snRNA):** Involved in the splicing of RNA transcripts.
- **MicroRNA (miRNA) and small interfering RNA (siRNA):** Involved in regulating gene expression by silencing specific mRNA molecules.

Comparison of DNA and RNA

Category	DNA	RNA
Sugar	2'-deoxyribose	Ribose
Bases	Adenine, Guanine, Cytosine, **Thymine**	Adenine, Guanine, Cytosine, **Uracil**
Structure	Double helix	Single-stranded (can fold into complex shapes)
Stability	Very stable	Less stable, more reactive
Primary Function	Long-term storage of genetic information	Gene expression (messenger, ribosomal, transfer RNA)
Location	Primarily in the nucleus (eukaryotes)	Synthesized in the nucleus; functions in cytoplasm and nucleus

THANK YOU